MIL-C-40077E
30 December 1985
SUPERSEDING
MIL-C-40077D
21 June 1971

MILITARY SPECIFICATION

CABINET, GASOLINE FIELD RANGE OUTFIT, MODEL 59, AND CRADLE, COOK POT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a metal cabinet and a cook pot cradle to be used with a burner unit for field cooking operations.

1.2 Classification. The cabinets and cradle shall be one of the following as specified (see 6.2):

Item I	-	Cabinet without cradle
Item II	-	Cabinet with cradle
Item III	-	Cradle, cook pot

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

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SPECIFICATIONS

FEDERAL

QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
QQ-A-250/8	-	Aluminum Alloy 5052, Plate and Sheet
QQ-S-741	-	Steel, Carbon: Structural Shapes, Plates, and Bars
QQ-S-763	-	Steel Bars, Shapes, and Forgings, Corosion-Resisting
QQ-S-766	-	Steel Plates, Sheets, and Strip Corrosion-Resisting
QQ-W-461	-	Wire, Steel, Carbon, (Round, Bare and Coated)
QQ-W-470	-	Wire, Steel, Carbon, Spring, Music
RR-C-271	-	Chains and Attachments, Welded and Weldless
MMM-A-260	-	Adhesive, Water-Resistant, (for Sealing Waterproofed Paper)
PPP-B-636	-	Boxes, Shipping Fiberboard
PPP-B-1055	-	Barrier Material, Waterproof, Flexible
PPP-T-60	-	Tape; Pressure Sensitive Adhesive, Waterproof, For Packaging

MILITARY

MIL-P-116	-	Preservation, Method of
MIL-C-11133	-	Crates, Shipping Wood, Open, Wirebound
MIL-T-12879	-	Treatments, Chemical, Prepaint and Corrosion Inhibitive, For Zinc Surfaces

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

5-11-71	-	Cabinet, Gasoline, Field Range Outfit; Cradle, Cook Pot
5-11-414	-	Assembly
5-11-415	-	Sections
5-11-416	-	Frame Assembly
5-11-417	-	Lid Assembly and details
5-11-418	-	Lid Details
5-11-419	-	Upper Door Assembly and Details
5-11-420	-	Upper Door Details
5-11-421	-	Upper Door Details

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5-11-424	-	Lifting Handle Assemblies and Miscellaneous Details
5-11-425	-	Lifting Handle Details
5-11-426	-	Outside Sheet and Inside Sheet Assemblies
5-11-731	-	Container for
5-11-910	-	Fire Unit Retainer Detail and Bracket Details

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting officer or as directed by the contracting activity.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

American Society of Mechanical Engineers

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 47	-	Malleable Iron Castings
A 108	-	Steel Bars, Carbon, Cold Finished, Standard Quality
A 153	-	Zinc Coating (Hot Dip) On Iron and Steel Hardware
A 269	-	Seamless and Welded Austenitic Stainless Steel for General Service
A 513	-	Electric-Resistant - Welded Carbon and Alloy Steel Mechanical Tubing
A 519	-	Seamless Carbon and Alloy Steel Mechanical Tubing
A 568	-	General Requirements for Steel, Carbon and High-Strength Low-Alloy Hot-Rolled Sheet and Cold-Rolled Sheet
A 575	-	Bars, steel, Merchant Quality M-Grades
B 633	-	Electrodeposited Coatings of Zinc on Iron and Steel
D 3951	-	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed end item complies with all the provisions of this document.

3.2.1 Aluminum sheet. Aluminum sheet shall conform to QQ-A-250/2 and QQ-A-250/8, as applicable.

3.2.2 Carbon steel.

3.2.2.1 Bar. Carbon steel bar shall conform to ASTM A 575 or ASTM A 108, as applicable.

3.2.2.2 Angle. Carbon steel angle shall conform to QQ-S-741.

3.2.2.3 Sheet and strip. Carbon steel sheet and strip shall be hot-rolled, finish conforming to ASTM A 568.

3.2.2.4 Tube. Carbon steel tube shall conform to ASTM A 513 or A 519.

3.2.2.5 Wire. Carbon steel wire for the latching pin shall conform to finish 5 of QQ-W-461.

3.2.3 Corrosion-resisting steel.

3.2.3.1 Bar. Corrosion-resisting steel bar shall be cold finished, flat form of QQ-S-763.

3.2.3.2 Sheet and strip. Corrosion-resisting steel sheet and strip of the types specified on the drawings shall conform to the applicable requirements of QQ-S-766.

3.2.3.3 Tube. Corrosion-resisting steel tube of the type specified on the drawing shall conform to ASTM A 269.

3.2.4 Castings. Malleable iron casting shall conform to ASTM A 47.

3.2.5 Chain. Chain shall conform to type I, grade C, class 5, style 1 of RR-C-271.

3.2.6 Wire, spring. Steel spring wire for the latching pin spring shall conform to QQ-W-470.

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3.3 Design and construction. The construction of all components and assemblies of the items covered herein shall conform to the requirements of this document and to the design, details, dimensions, material classification, and sizes thereof shown on applicable drawings listed in 2.1.

3.3.1 Handles, lifting. The left and right lifting handles, shall be fabricated from tube specified in 3.2.2.4. When welded tube is used, the weldment seam shall be on the inside when assembled on the cabinet (not exposed when handles are in lowered position).

3.3.2 Brackets and connectors. The brackets and connectors, items 13-2, 13-8, 14-2, 16, and 17 shown on Drawing 5-11-424 shall be cast from material specified in 3.2.4 or fabricated by weldment, as an alternate construction, provided all parts are interchangeable, ductile, and equal to or better in strength. A sample of the proposed alternate construction part shall be furnished the contracting officer for approval prior to assembly on the cabinet.

3.4 Finish.

3.4.1 Surface preparation. All metal surfaces shall be cleaned and degreased in accordance with standard commercial practice before any preservative finish is applied.

3.4.2 Zinc coating. Except when otherwise specified, all steel parts, except parts fabricated of corrosion-resisting steel, shall be zinc coated by either the electrolytic or hot dip galvanizing process as specified on the drawing. Zinc coated parts that require bending shall be bent in such a manner as to minimize cracking and flaking. All zinc coated areas exhibiting cracking or flaking shall be repaired to a finish comparable to the finish before bending and shall show no visible evidence of finish damage. Discoloration due to chromate conversion coatings is permissible.

3.4.2.1 Electrolytic coating. Electrolytic zinc coating shall conform to type II of ASTM B 633, with a minimum thickness of 13 micrometers.

3.4.2.2 Hot-dip galvanizing. Hot-dip galvanized coating shall conform to ASTM A 153. Such coating shall be chromate treated in accordance with type II of MIL-T-12879.

3.5 Marking. Marking shall be as shown on Drawing 5-11-71 and 5-11-421 as applicable, and in accordance with MIL-STD-130.

3.6 Workmanship. The cabinet corners shall be square within 1/8 inch. The door and lid shall be in alignment. Moving parts shall not bind. The cabinet shall be clean and free of dents, cuts, tears, and tool marks. All metal used in the fabrication of the cabinet shall be free of kinks, sharp bends, and

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slivers. Shearing shall be done neatly and accurately. Bends shall be made accurately to provide uniformity of size and shape. All pin, stud, and screw holes shall be accurately drilled and all burrs shall be removed.

3.6.1 Welders. Unless otherwise specified, before assigning any welder to work, the contractor shall provide the Government inspection service with the names of the welders to be employed in the work, together with certification that each of the welders has passed a qualification test as prescribed by the American Society of Mechanical Engineers for the type of welding operation to be performed, and that such qualification is effective as defined by the particular code.

3.6.2 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be sound, continuous, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds. Frame welds shall be ground flush to the frame surface prior to assembly of steel sheets.

3.6.3 Riveting. Holes for rivets shall be accurately punched or drilled. Rivets shall be tight, fully peened with heads in full contact with the surface of the metal. Rivets, where possible, shall be driven from inside the cabinet and peened on the outside.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 6.2) is shall be examined for the defects specified in 4.4.2.1 and 4.4.2.2. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 In-process inspection. Examination shall be made of the following operations to establish conformance with document requirements. Whenever non-conformance is noted, correction shall be made to the applicable operation and all items processed.

a. Preparation of metal surfaces, as applicable (see 3.4.1).
b. Coating and retouching of cracked and flaked areas (see 3.4.2).
c. Metal fabrication (see 3.6).
d. Welding and preparation of surfaces to be welded (see 3.6.2).
e. Riveting (see 3.6.3).

4.4.2 End item inspection. For cabinets, the inspection lot shall be all cabinets offered for inspection at one time. The sample unit shall be one completely fabricated cabinet. For cradles, the inspection lot shall be all cradles offered for inspection at one time. The sample unit shall be one completely fabricated cradle.

4.4.2.1 Visual examination. The end items shall be examined for the defects listed in table I, as applicable. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish (coating)	Not smooth, lumps or blisters		X
	Coating cracked, area of no coating	X	
Construction and workmanship	Kinks or sharp bends	X	
	Sharp burrs	X	
	Door and lid not in alignment	X	
	Moving parts bind	X	
	Dents, cuts, tears, or toolmarks	X	
Welding	Missing, incomplete, porous, or cracked	X	
	Scale or flux deposits not removed.		X
Rivets	Missing	X	
	Heads not in full contact with surface of adjacent metal	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Marking for identification	Missing, incomplete, not legible, not specified type or size		X

4.4.2.2 Dimensional examination. The cabinet and cradle, as applicable, shall be examined for compliance with dimensions specified. Any dimension that is not within the specified tolerance shall be classified as a defect. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.3 Packaging inspection. An inspection shall be made to determine that preservation, packing, and marking comply with section 5. Defects shall be as indicated in table II. The lot size shall be the number of shipping containers in the inspection lot. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE II. Packaging defects

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Components missing or damaged
Workmanship	Inadequate application of components, such as incomplete closure of shrouds. Bulged or distorted container
Preservation	Improperly applied or missing

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservation. All uncoated ferrous metal surfaces of the cabinet shall be cleaned process C-1, thoroughly dried, and then coated with type P-14

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preservative in accordance with MIL-P-116. When furnished separately, the cook pot cradle shall be preserved method I of MIL-P-116. The cleaning process and preservative shall be as specified herein.

5.1.1.2 Unit packing. Each cook pot cradle, when furnished with cabinet, shall be placed inside the cabinet and securely wired at four points to the rails with minimum 0.07-inch diameter annealed wire to prevent movement while in transit. The cabinet lid shall be closed and wired in position with wire specified herein. The wire shall pass through the lid handle and upper front door latch. The shutters in the upper door shall be closed and taped in position with tape conforming to PPP-T-60. The pin latch shall be secured with tape specified herein.

5.1.2 Commercial. The cook pot cradle shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Cabinet, or cabinet with cook pot cradle. Each cabinet, or each cabinet with cook pot cradle when required, preserved as specified in 5.1, shall be packed in a wire bound crate constructed in accordance with Drawing 5-11-731. The cabinet shall be secured to the base of the crate with four 3/8-inch carriage bolts. Lockwashers shall be used to prevent loosening of the nuts. The bolts shall extend through the battens, the bottom slats, and up through the holes in the unit provided for bolting purposes. Each cabinet shall be covered with a full height shroud fabricated of waterproof barrier material conforming to PPP-B-1055. All seams of the shroud shall be sealed with a 3/4-inch minimum width continuous seam of water-resistant adhesive conforming to MMM-A-260. The crate body or mat shall be folded around the cabinet so that the outer cleats of the mat engage in the channels of the top and base of the crate. Nailing of the top and bottom to the mat shall not be required. Crate shall be closed by securing the looped closures in accordance with the appendix of MIL-C-11133, except that a hammer shall not be used.

5.2.1.2 Cook pot cradle. Each cook pot cradle, when furnished separately, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B packing.

5.2.2.1 Cabinets, or cabinets with cook pot cradle. Cabinets or cabinets with cook pot cradles, preserved as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the waterproof barrier shroud shall not be required.

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5.2.2.2 Cook pot cradle. Each cook pot cradle, when furnished separately, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic, variety optional, grade 200 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.2.1 Weather-resistant containers. When specified (see 6.2), the shipping container shall conform to V3c, V3s, or V4s fabricated and closed in accordance with PPP-B-636.

5.2.3 Commercial, packing. The cook pot cradle preserved and packaged as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The cabinet and cradle cook pot are to be used with the burner unit M2 covered by MIL-B-40098 for preparing food in the field.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Item required (see 1.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.2.1)

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample consisting of one unit. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Preparing activity:

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-40077E	Cabinet, Gasoline Field Range Outfit, Model 59 and Cradle, Cook Pot

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84

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